

**Public Comment in response to the Request for Information (RFI) on the
Development of an Artificial Intelligence (AI) Action Plan**

FlexHEGs: Hardware-Enabled Security to Support American AI Leadership

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Addressed to:

Faisal D'Souza, NCO
Office of Science and Technology Policy
Executive Office of the President
2415 Eisenhower Avenue
Alexandria, VA 22314

Addressed from:

Ethan Ashkie
Chief Executive Officer
Survival and Flourishing .Com, PBC
2991 Sacramento St #516
Berkeley, CA 94702

Submitted by email to 

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Introduction

We strongly support the Administration's effort to define priority policy actions needed to sustain and enhance America's AI dominance. In this submission, we outline a highly promising computer chip security mechanism called a **Flexible Hardware-Enabled Guarantee (FlexHEG)**, which could solve critical security challenges while cementing American leadership in AI development.

Recent advances in AI capabilities suggest that even more powerful systems will emerge over the coming years. The Administration faces dual challenges: accelerating American innovation while ensuring that powerful AI technologies are not used by foreign adversaries to threaten US national security interests. A hardware-based approach that provides verifiable security measures while allowing unfettered domestic innovation would address both challenges simultaneously.

FlexHEG technology is thus ideal for advancing American interests: a security system that would be integrated into AI chips, combining a tamper-proof enclosure with a secure processor to enable trustworthy, privacy-preserving verification and enforcement of claims about how those chips are used. This technology would allow the US to extend its technological leadership while enforcing accountable development practices.

What Are FlexHEGs?

FlexHEG systems combine two key components:

1. **Secure processor:** This component measures the AI chip's operations and can enforce security policies or verify claims about chip usage. Unlike typical cybersecurity approaches, the secure processor provides hardware-level verification that cannot be circumvented by software manipulation.
2. **Tamper-proof enclosure:** This physical protection prevents unauthorized access to the hardware, ensuring that security measures cannot be bypassed through physical modifications. If tampering is detected, the system can trigger countermeasures ranging from simple alerts to complete self-disablement.

These components work together to create a technical foundation for verifiable claims about chip usage while maintaining privacy. Most importantly, FlexHEGs would be designed to be fully open-source and auditable, making it suitable for deployment in contexts where trust is limited—such as export-controlled chips sent to lower trust jurisdictions or security-critical applications.

The secure processor would operate like a black box recorder for AI systems—monitoring what happens without unnecessarily restricting operations for legitimate users. For exports or high-risk deployments, additional restrictions could be configured as needed.

Early prototypes of FlexHEGs have already been demonstrated. Initial scoping work has begun, supported by philanthropic funding. An interim report has been prepared¹, and an updated comprehensive report will be available to the Administration shortly.

Enhancing American National Security

Preventing Theft and Unauthorized Access to AI Systems

The theft of American intellectual property and AI model weights represents an increasingly serious threat to our technological advantage. Losses of IP aren't just economic—they erode America's technological edge by providing adversaries immediate access to capabilities that may have required billions in R&D investment to create.

FlexHEGs would introduce strong hardware-based protections that go beyond software security measures:

- **Tamper-responsive enclosure:** Advanced physical protection of chips, with sensors to detect tampering attempts and trigger deletion of data before it can be exfiltrated.
- **Model weight encryption:** In-line encryption to protect all data that leaves the secure enclosure.

These protections would prove particularly valuable for advanced AI models that represent the product of computational resources strategically denied to adversarial nations through export controls. A single theft of a frontier AI model could significantly undermine US strategic advantage by providing adversaries immediate access to refined intellectual property derived from restricted computational resources.

Securing Exports While Expanding Markets

A key advantage of FlexHEGs is their ability to maintain security controls even after AI chips have been shipped to lower trust jurisdictions. Without hardware-based security measures, the Administration faces a stark choice: either restrict exports of advanced AI chips (severely limiting market access for American chip manufacturers) or accept significant security risks. FlexHEG systems would enable:

¹ https://yoshuabengio.org/wp-content/uploads/2024/09/FlexHEG-Interim-Report_2024.pdf

- **Controlled deployment in lower trust environments:** American AI models could be deployed internationally, with FlexHEGs providing an additional layer of security on top of conventional defenses for protecting model weights
- **Verification of operating restrictions:** Exports could be configured to only permit approved uses, with hardware-level verification
- **Remote disablement:** Configuration options could allow US authorities to remotely disable chips if they are diverted to unauthorized uses or locations

This approach mirrors successful precedents such as Trusted Platform Modules (TPM) which enable systems to operate securely in adversarial environments, and Hardware Security Modules (HSM) which are used by the American cryptography industry to protect sensitive operations.

Bolstering American Technology Leadership

Maintaining Compute Advantage Through Hardware Security

Computational capacity represents a crucial advantage for American AI development, in particular the development of the most advanced frontier AI models. Unlike data or algorithms, compute is a relatively controllable resource, making it an effective point of leverage for national technology policy. Current export controls aim to preserve America's current advantage by denying adversaries access to advanced AI chips, but this approach faces growing challenges from smuggling and the development of alternative chips.

FlexHEGs provide a more effective approach to maintaining compute advantage:

- **Verification of chip usage:** Hardware-level monitoring of how chips are being utilized could ensure compliance with export restrictions
- **Prevention of cluster aggregation:** FlexHEGs could prevent unauthorized clustering of chips into larger systems capable of training the most advanced models
- **Quantitative limits:** Systems could be configured to enforce specific compute thresholds, preventing training runs above certain sizes

By creating technical constraints on compute aggregation that are enforceable through hardware, FlexHEGs could grant foreign users sufficient compute access for legitimate commercial applications while preventing foreign adversaries from assembling the large compute clusters required for training frontier AI models.

Developing Technical Standards

Hardware security standards are particularly effective when they originate from a position of market dominance. By developing and implementing FlexHEGs, the US would establish de facto technical standards that favor American interests.

This approach has several advantages:

- **First-mover benefit:** Early establishment of standards tends to create path dependency in subsequent technology development
- **Market leverage:** American companies' dominant position in AI chips provides significant leverage for standard-setting
- **Technical control points:** Hardware-based standards create lasting control points that are difficult for competitors to circumvent

The Administration should encourage American companies to develop common standards for hardware security in AI chips, potentially using NIST as a coordinating body. These standards could then be incorporated into export control requirements, creating strong incentives for adoption.

Enabling Trusted AI Development Without Sacrificing Innovation

Recent proposals for AI safety have raised concerns about their potential to hamper American innovation through excessive reporting requirements or restrictions. FlexHEG offers an alternative approach: verifiable security and safety claims without unnecessary disclosure or restriction.

For example, rather than requiring companies to publicly disclose training details that might compromise their competitive advantage, FlexHEGs would enable them to make verifiable safety claims without revealing proprietary information.

This approach aligns with the Administration's goals of removing unnecessary barriers to American leadership in AI while ensuring appropriate safety measures are in place.

Immediate Security Gains Through Existing Hardware

While comprehensive FlexHEG systems represent the future of AI hardware security, significant security enhancements are achievable in the near term through firmware updates to existing AI chips combined with software-based approaches. Current hardware already contains security features that could be leveraged immediately to address pressing national security concerns:

- **Ping-based geolocation:** Firmware updates could enable verification that chips remain in authorized jurisdictions
- **Offline licensing enforcement:** Automatic disabling of chips detected outside their authorized locations

- **Enhanced model protection:** Strengthened security measures to prevent theft of valuable AI models
- **Verifiable evaluations:** Privacy-preserving verification of AI model capabilities and properties

Manufacturers like NVIDIA could likely develop and deploy these capabilities to existing H100s and other advanced AI chips in a matter of months, through a combination of firmware updates and software-based approaches. The Administration could encourage or incentivize this course of action through various means. These capabilities would build upon existing hardware security, which would be challenging for adversaries to circumvent, providing immediate security benefits while more comprehensive FlexHEG systems are being developed.

It would be a mistake for the Administration to wait until FlexHEG technology was fully developed before pursuing increased hardware security. The full FlexHEG system will extend these capabilities, introducing purpose-built hardware with both greater security against expensive state-level attacks and the flexibility to address evolving governance requirements.

Implementation Roadmap

We recommend the Administration pursue a phased approach to developing and implementing FlexHEG technology. This approach acknowledges both the urgency of addressing AI security challenges and the technical complexity involved in developing robust hardware solutions.

Phase 1: Initial R&D and Pilot Deployments (2025-2026)

The Administration should immediately initiate research into secure processors and tamper-proof enclosures tailored for AI chips. Early versions could focus on verification capabilities, with enforcement mechanisms to be added in later iterations.

Actions should include:

- **DARPA research program:** Launch a dedicated program for developing FlexHEG technology, including tamper-responsive enclosures, auditable processor IP blocks, and algorithms to automatically check for rules
- **NIST standards development:** Direct NIST to develop preliminary standards for hardware security in AI chips
- **Pilot deployments:** Fund initial deployments in national security applications
- **Industry engagement:** Establish a working group with leading AI chip manufacturers to ensure compatibility with commercial technology roadmaps

These initial efforts would focus on establishing technical feasibility and creating reference implementations that could be refined in subsequent phases.

Phase 2: Commercial Integration and Export Applications (2026-2027)

Building on the results of Phase 1, the Administration should focus on integrating FlexHEG technology into commercial AI chips and applying it to secure exports of advanced computing technology.

Actions should include:

- **Export control integration:** Modify export control requirements to favor FlexHEG-equipped chips
- **Commercial incentives:** Provide expedited planning approval with faster security auditing for FlexHEG-equipped systems
- **Retrofitting program:** Develop capabilities for adding FlexHEG protections to existing chips, allowing rapid implementation

By the end of Phase 2, FlexHEG technology should be sufficiently mature for deployment in export-controlled contexts, allowing more flexible export policies while maintaining security.

Phase 3: Widespread Adoption and Sophisticated Applications (2027-2028)

In the final phase, the Administration should focus on driving widespread adoption of FlexHEG technology and leveraging it for increasingly sophisticated security applications.

Actions should include:

- **Regulatory requirements:** Institute requirements for FlexHEG verification in high-risk AI applications
- **Advanced security features:** Develop more sophisticated security features based on lessons from earlier deployments
- **Integration with AI safety frameworks:** Connect hardware verification to broader AI safety initiatives
- **Enforcement infrastructure:** Establish the necessary infrastructure for monitoring compliance and responding to violations

By this phase, FlexHEGs should become a standard feature of advanced AI chips, providing robust security for American AI technology regardless of where it is deployed.

Recommended Policy Actions

To advance this technology and its strategic benefits, we recommend the following policy actions for inclusion in the AI Action Plan:

1. Establish a National Security Presidential Memorandum on AI Hardware Security

The President should issue a National Security Presidential Memorandum directing relevant agencies to prioritize the development and deployment of hardware security measures for AI chips. This memorandum should:

- Instruct the Department of Commerce to develop export control provisions favoring FlexHEG-equipped hardware
- Direct the Department of Defense and Intelligence Community to develop capabilities for evaluating hardware security in AI systems
- Establish a cross-agency task force to coordinate FlexHEG development and deployment
- Direct NIST to develop technical standards for secure AI hardware

A Presidential directive would create the necessary sense of urgency and ensure coordinated action across relevant agencies.

2. Direct Implementation of Near-Term Hardware Security Measures

The Administration should immediately direct the implementation of firmware- and software-based security measures for existing AI chips:

- Partner with private chip manufacturers to develop and deploy firmware and software updates that enable basic security features on existing hardware, using federal procurement leverage and voluntary industry collaboration
- Establish a fast-track working group between the Department of Commerce, NIST, and leading chip manufacturers to define technical requirements for these updates
- Create incentives for voluntary adoption by domestic users while requiring implementation for exported hardware
- Consider invoking the Defense Production Act if absolutely necessary in order to accelerate implementation

These near-term measures would provide immediate national security benefits while more comprehensive FlexHEG systems are being developed, creating a bridge between current vulnerabilities and the robust security infrastructure outlined in this proposal.

3. Fund a Major DARPA Research Program

The Administration should direct DARPA to establish a major research program focused on developing secure processor technology and tamper-proof enclosures for AI chips. This program should receive initial funding of at least \$100 million and focus on:

- Open-source secure processor designs optimized for AI applications
- Advanced tamper-detection and response mechanisms
- Algorithms for automatically checking whether chip usage is compliant
- Methods for retrofitting security to existing hardware

DARPA's experience with high-risk, high-reward research makes it ideally suited to drive the initial development of FlexHEG technology.

4. Create Export Control Incentives

The Department of Commerce should modify export control regulations to provide streamlined approval processes for FlexHEG-equipped AI chips. These modifications should:

- Establish a technical certification process for FlexHEG compliance
- Create expedited approval pathways for certified systems
- Allow broader deployment of certified systems to Tier 2 countries under the Framework for AI Diffusion
- Develop mechanisms for verifying ongoing compliance through hardware attestation

These incentives would drive commercial adoption while advancing national security objectives.

5. Establish Procurement Preferences

The Administration should direct federal agencies to establish procurement preferences for AI systems equipped with FlexHEG technology. This would:

- Create immediate market demand for FlexHEG-equipped systems
- Provide real-world testing and refinement opportunities
- Demonstrate federal commitment to the technology
- Improve security for government AI deployments

Federal procurement represents a powerful tool for driving technology adoption and should be leveraged to accelerate FlexHEG deployment.

6. Direct NIST to Develop Technical Standards

NIST should be directed to develop comprehensive technical standards for hardware security in AI chips, including:

- Reference designs for secure processors
- Testing and certification protocols
- Attestation formats and procedures
- Requirements for tamper-evident and tamper-responsive features

These standards would provide a foundation for industry implementation and regulatory requirements.

Conclusion

The continued advancement of AI capabilities presents both enormous opportunities and significant challenges for American leadership. By developing and deploying FlexHEG technology, the Administration can address critical security concerns while promoting continued innovation. This approach offers several key advantages:

- It maintains American technological leadership by securing AI technology against theft and misuse
- It enables more flexible export policies by providing hardware-level security for chips deployed internationally
- It creates a technical foundation for addressing emerging security threats posed by advanced AI capabilities
- It establishes American dominance in the critical domain of AI hardware security

The window of opportunity to establish technical standards in this domain may be relatively brief. As AI capabilities continue to advance and foreign competitors develop alternative chip technologies, the ability to implement effective hardware security measures may become more limited. By acting decisively now, the Administration can secure lasting advantages for American AI technology.

The development of FlexHEGs does not require fundamental scientific breakthroughs—all the necessary technologies exist today in various forms. What's needed is focused development, integration, and deployment, driven by clear policy directives and adequate resources. By including FlexHEG development in the AI Action Plan, the Administration would take a significant step toward ensuring continued American dominance in AI.

Addendum

An updated report detailing the latest findings of FlexHEG development is currently being written and can be made available to the Administration by request.